Md Hashibul Hossain

Junior Research Scientist, CredoSense (Dhaka) · MSc (Thesis) in Forestry, With Distinction

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Research profile

Forestry-trained environmental scientist working on how ecosystems respond to environmental stress through two complementary paths. In the wet lab I study fungal heavy-metal tolerance and bioremediation, plant pigments and nutrition, and plant pathogens, using culture assays, antioxidant enzyme profiling and ICP-OES. In the dry lab I use satellite remote sensing, drone photogrammetry and computer vision to map vegetation, land cover and contamination risk, and I write reproducible R and Python analyses. MSc (Thesis) in Forestry With Distinction; currently Junior Research Scientist at CredoSense, Dhaka. Seeking a PhD that links lab-measured mechanisms with landscape-scale environmental and ecological questions.

Research interests

Environmental microbiology · fungi–heavy metal interactions and mycoremediation · plant pathology and pathogen detection · plant physiology and nutrient chemistry · mycorrhizal ecology and soil carbon · remote sensing of vegetation and land cover · drone photogrammetry · computer vision for biology · urban green space and landscape ecology · fungal–plant–animal interactions in tropical forest restoration

Education

Session 2022 – 2024

MSc (Thesis) in Forestry

Shahjalal University of Science & Technology (SUST), Sylhet

CGPA 3.80 / 4.00 · With Distinction · result published Feb 2026

- Classes began Mar 2024; thesis defended Jan 2026 (schedule delayed by the 2024 campus closures)
- Thesis: tolerance, bioaccumulation and bioremediation of heavy metals in water and soil by Aspergillus niger and Trichoderma asperellum, with screening of metallo-tolerant landfill isolates
- Supervisor: Dr Mohammed Masum Ul Haque · Co-supervisor: Prof. Romel Ahmed
- Data Analysis and Visualization in R (A+); Dissertation and Thesis both A+

Session 2018 – 2022

BSc (Honours) in Forestry

Shahjalal University of Science & Technology (SUST), Sylhet

CGPA 3.43 / 4.00 · result published May 2024

- Final examinations held Dec 2023 (schedule delayed by COVID-19)
- Final-year project (2022 – 2023): in-vitro assessment of heavy-metal tolerance of selected fungi
- Coursework: Remote Sensing and GIS, Mycology and Forest Pathology, Genetics and Tree Improvement, Research Methodology

Research experience

Jul 2026 – present

Junior Research Scientist

CredoSense (Canada-based) · Dhaka office, Bangladesh

- Machine-learning and computer-vision models that detect pathogenic fungi from microscopic spore images
- Object-detection models for greenhouse pests: curation, annotation and evaluation
- Plant pigment extraction (chlorophyll, carotenoids) and spectrophotometry
- Plant-tissue digestion and macro/micronutrient analysis by ICP-OES

Jan – Dec 2025

Research Associate

Dept. of Forestry & Environmental Science, SUST

- Designed and led a fungal bioremediation project, including landfill sampling and isolation
- Isolated fungi from resin-forming agarwood trees
- Operated ICP-OES independently; UV-Vis colorimetric and enzyme assays
- Statistics in R; presented at international conferences; mentored junior researchers

Jan – Dec 2024

Research Assistant

Dept. of Forestry & Environmental Science, SUST

- Experiment design for plant–pathogen interaction and metal-stress studies
- Ran seed-susceptibility tests against fungi
- Literature review, laboratory protocols, controlled sample handling

Jan 2024

Intern and Field Researcher

USAID Bangladesh Ecosystems / Protibesh Activity · Khadimnagar National Park · Supervisor: Md. Modinul Ahsan, PhD

- Compulsory internship in protected-area biodiversity management
- Mixed-methods study of co-management office bearers' and People's Forum members' knowledge of the Protected Area Management Rules, 2017
- 22 respondents from 10 villages (purposive sampling, structured questionnaire, field observation); analysis in R 4.3

Fieldwork

11 sites

Agarwood plantation survey

Surveyed 11 plantations alone in about 10 days; soil sampling and tree inventory in plots covering 5% of plantation area

Camera traps

Camera-trap study

Chose camera-trap sites for nocturnal carnivores from food-resource distribution; mapped and analysed detections in GIS

Landfill

Landfill rhizosphere sampling

GIS-selected site; 7 root-zone soil and 3 water samples; aseptic collection, cold transport, –80 °C storage

Forest plots

Vegetation plot survey, Satchari National Park

Laid plots, identified the plants and counted all timber species in them

Forest types

Forest types worked in

Hill forest (Sylhet, Chittagong Hill Tracts), sal forest (Madhupur), mangrove (Sundarbans)

Selected research projects

2024 – 2026

Heavy-metal tolerance, bioaccumulation and bioremediation by filamentous fungi

MSc thesis · SUST · Supervisors: Dr M. M. U. Haque, Prof. R. Ahmed

Dose–response screening of Aspergillus niger and Trichoderma asperellum against As(V), Cd(II), Cr(VI) and Pb(II), with enzyme profiling and removal tests in water and soil microcosms.

2024 – 2025

Metallo-tolerant fungi from the Lalmatia waste dump

MSc field discovery pipeline · Sylhet

GIS-chosen sampling site, 10 field samples (7 soil, 3 water), bioavailable metals by ICP-OES, and isolation of 27 morphotypes by serial dilution.

- ITS-based identification in progress
- Factorial assays of up to 210 plates in one day (7 fungi × 2 metals × 5 concentrations × 3 replicates)
- Prepared and counted spore suspensions to inoculate soil in an ongoing study of the 7 isolates

2025 – present

Fungal associations of resin-forming agarwood

Research Associate · SUST

Isolation of fungi from resin-forming Aquilaria trees to study fungus–plant associations.

- Isolated fungi from resinous wood and helped inoculate living agarwood trees (results pending)

2024

Mycorrhizal fungi of dipterocarp forest trees

Culture work · SUST

Cultured fungi from fruit bodies associated with telsur (*Hopea odorata*) and garjan (*Dipterocarpus*) on MMN medium.

2024 – 2025

Multi-sensor GIS site selection for contamination sampling

MSc thesis · ArcMap 10.8 · Google Earth Engine · R

Multi-criteria suitability model fusing atmospheric, thermal, vegetation, terrain and hydrology layers on a common 500 m grid over a 10-year window, with sensitivity analysis.

2022 – 2023

In-vitro heavy-metal tolerance of three fungi

BSc project · SUST

A. niger, R. stolonifer and T. asperellum on PDA amended with Pb, Hg and As at 100–500 ppm, growth measured at 24, 48 and 72 h.

Dec 2024 – Feb 2025

YOLO11s leaf detector with TensorFlow.js export

Python & Data Science training · IICT-SUST

Transfer-learned YOLO11s for 2-class leaf detection (1,107 train / 46 val images, 100 epochs), validated and exported for in-browser inference.

2023 **Land-cover change, Karnaphuli Range (Kaptai)**
Coursework · SUST
Satellite image analysis of land-use and land-cover change across a ~560 ha forest range.

Research software & open code

Python
AGPL-3.0
Dīvi: drone mapping and canopy analysis for researchers
github.com/Mraz11/Divi
Offline-first toolkit for consumer RGB drones (formerly Avispria): terrain-aware DJI Fly mission planning, photo QA, WebODM orthophoto/DSM processing, canopy height, cover, gaps and tree-crown analysis. 17.9k lines of Python, 1,053 automated tests. Early version; one test flight (DJI Lito X1, 79 photos), results not yet validated.

R
MIT
Estimating stress tolerance: dose–response modelling done properly
github.com/Mraz11/dose-response-ec50
LL.4 reproduces the published ryegrass ED50 exactly (3.058, 95% CI 2.671–3.445). On labelled synthetic data, RMSE of log-EC50 falls from 0.21 to 0.055 across 12 designs, 1,000 simulations each.

R
MIT
Natural background vs anthropogenic enrichment of toxic elements in US soils
github.com/Mraz11/soil-metal-enrichment
Zn, Cd and Ni show land-cover-linked enrichment robust to the normaliser. Pb's strongest Al-based signal (p = 6.2e-13) did not replicate with Ti (p = 0.68) and is reported as unconfirmed.

R
MIT
Climate and ecosystem controls on global soil respiration
github.com/Mraz11/climate-controls-soil-respiration
MAT × biome interaction is supported by LRT (p = 0.036; 0.0019 after removing 28 influential points) but does not improve prediction: exact leave-one-study-out CV over 1,247 studies (2,494 refits) gives predictive $R^2 \approx 0.11$.

Python
MIT
Metal-resistance genes in bacteria from contaminated vs reference environments
github.com/Mraz11/metal-resistance-genes-environment
Contaminated genomes carry about 4% more metal-resistance genes per 1,000 proteins (rate ratio 1.039, 95% CI 0.996–1.083). Significant with HC1-robust errors (p = 0.030) but not once clustered by sampling site (p = 0.119) or clade (p = 0.076): 84 genomes cannot confirm the effect.

R
MIT
Dose–time metallotolerance screening in fungi
github.com/Mraz11/dose-time-metallotolerance-fungi
R pipeline producing Figures 1–8 for A. niger, T. asperellum and R. stolonifer under As, Pb and Hg: tolerance index, growth rate, Tukey contrasts, TI heatmap, LL.4 curves, IC50 with bootstrap CI and censoring.

Publications & presentations

MANUSCRIPTS IN REVIEW

1. **Hossain, M.H.**, Haque, M.M.U., Ahmed, R., & Rahman, M.A. *Integrated screening of metal tolerance and short-contact dry-biomass biosorption reveals metal-specific performance in filamentous fungi.* In review
2. Aziz, F.B.A., Sultana, R., Masum, K.M., & **Hossain, M.H.** *Potential urban green space suitability appraisal using GIS-based multi-criteria analysis: A case study of North-eastern Sylhet City, Bangladesh.* In review
3. Rekha, P.R., Ghose, A., **Hossain, M.H.**, & Masum, K.M. *Urban green spaces as carnivore refugia: A camera-trap study in a biodiversity-rich university campus in Bangladesh.* In review

CONFERENCE PAPER

4. **Hossain, M.H.**, Ahmed, R., & Haque, M.M.U. (2025). *Heavy metal tolerance by fungi: A step toward mycoremediation strategy for contaminated soils.* International Conference and Bioscience Carnival, Rangamati Science and Technology University. doi:10.13140/RG.2.2.29154.03525

POSTERS

5. **Hossain, M.H.** (2023). *Mycorrhizal role of carbon sequestration in forest soil.* Contemporary Trends in Forestry Research seminar, SUST.
6. **Hossain, M.H.** (2023). *Bioaccumulation of heavy metals by fungi and its possible effects on the environment.* International Day of Forests, SUST.

REPORTS (NON-PEER-REVIEWED)

7. **Hossain, M.H.**, & Eva, N.J. (2024). *Knowledge of office bearers and People's Forum members of Khadimnagar National Park about the Protected Area Management Rules, 2017.* BSc internship report, USAID Ecosystems / Protibesh Activity.
8. **Hossain, M.H.** (2023). *Image analysis of Karnaphuli Range.* Coursework report, SUST.

Skills

Wet lab & analytical chemistry
ICP-OES · Microwave digestion · UV-Vis · Pigment extraction · CAT · SOD · GR assays · Aseptic culture · Serial dilution · Cryopreservation · Bioremediation microcosms · Plant pathology

Molecular
DNA extraction (fungi, bacteria) · PCR: ITS and 18S markers · BLAST · Phylogenetic analysis · FastQC on NCBI sequence data · ITS identification of DMFT1–7 (in progress) · Comparative genomics: DIAMOND, Mash

Statistics & code
R · Python · pandas · ggplot2 / seaborn · LL.4 dose–response · Bootstrap CI · Mixed models (lme4) · Cross-validation · ANOVA / Tukey · PCA · NMDS · renv · uv · Make · pytest · Git / GitHub

GIS & remote sensing
Google Earth Engine · ArcMap 10.8 · MCDA · Drone photogrammetry (WebODM) · DJI WPML · Sentinel-2 / 5P · Landsat LST · MAIAC AOD · VIIRS · SRTM · Land-cover change · i-Tree

Machine learning & vision
YOLO11 / Ultralytics · Transfer learning · Roboflow · Annotation · TensorFlow.js export · ImageJ · Microscopy image analysis

Field & social-science methods
Field sampling · Structured questionnaires · Purposive sampling · Stakeholder interviews · Mixed-methods analysis · Protected-area co-management

Training & certification

Jun 2025	**WEkEO for Climate Change** · Copernicus & Mercator Ocean International
Apr 2025	**Earth Observation for Climate Adaptation and Resilience in South Asia** · AIRCAS (CAS), Jagannath University, BRAC University C3ER
Feb 2025	**Core Techniques in Molecular Biology and NGS (5 days)** · Central Dogma Lab, Invent Technologies
Dec 2024 – Feb 2025	**Python Programming and Basic Data Science (80 h)** · IICT-SUST / Bangladesh Computer Council
Feb 2025	**Understanding Climate Change and Climate Finance** · PKSF / Green Climate Fund
Sep 2024	**ICP-OES (8 days)** · Invent Technologies at SUST
Sep 2023	**Bioinformatics Bootcamp 2.0** · CHIRAL Bangladesh
Jul 2023	**Urban Tree Inventory using i-Tree** · USFS International Programs

Leadership & service

Mar 2023 – Mar 2024	**G Studio Wing Chief** · Green Explore Society, SUST (9th Executive Committee)
Mar 2022 – Mar 2023	**Assistant Publication Secretary** · Green Explore Society, SUST (8th Executive Committee)
2025	**Mentor to junior researchers** · Research Associate role, Dept. of Forestry & Environmental Science, SUST

Awards & honours

2026	MSc awarded With Distinction
2025	National Eco Olympiad — National Round
2018	Bangladesh Physics Olympiad — Regional round
2017 – 2018	Bangladesh Chemistry Olympiad — Preliminary rounds (7th, 8th)
2017	Bangladesh Mathematical Olympiad — Regional round
2017	Bangladesh Botany Olympiad — Participant

Languages

English	IELTS Academic 7.5 overall (CEFR C1), Sep 2026 · Listening 8.0, Reading 8.5, Writing 6.5, Speaking 6.5
Bengali	Native

Referees

Dr Mohammed Masum Ul Haque
Associate Professor, Forestry & Environmental Science, SUST
BSc and MSc supervisor

Prof. Romel Ahmed
Professor, Forestry & Environmental Science, SUST
MSc co-supervisor

Prof. Kazi Mohammad Masum
Professor, Forestry & Environmental Science, SUST
Collaborator on the camera-trap manuscript and another ongoing paper

Contact details, and an additional referee from CredoSense, available on request.